

Администрирование сетевых подсистем

Лабораторная работа №3 — Настройка DHCP-сервера и динамических обновлений DNS

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Цели и задачи работы

Цель лабораторной работы

Приобрести практические навыки по установке и конфигурированию DHCP-сервера и настройке динамического обновления DNS-зон (DDNS).

Настройка DHCP-сервера

Базовая конфигурация Kea DHCP

```
root@server:~  
[vagrant@server ~]$ sudo -i  
root@server ~]# dnf -y install kea  
Last metadata expiration check: 0:16:46 ago on Wed 04 Mar 2026 03:29:26 PM UTC.  
Dependencies resolved.  
=====
```

Package	Arch	Version	Repository	Size
---------	------	---------	------------	------

```
=====
```

Installing:

kea	x86_64	2.6.4-1.el9	epel	1.3 M
-----	--------	-------------	------	-------

Installing dependencies:

kea-libs	x86_64	2.6.4-1.el9	epel	3.0 M
log4cplus	x86_64	2.0.5-15.el9	epel	337 k
mariadb-connector-c	x86_64	3.2.6-1.el9_0	appstream	195 k
mariadb-connector-c-config	noarch	3.2.6-1.el9_0	appstream	9.8 k
postgresql-private-libs	x86_64	13.23-1.el9_7	appstream	140 k

Transaction Summary

```
=====
```

Install 6 Packages

Total download size: 5.0 M
Installed size: 18 M
Downloading Packages:

(1/6): log4cplus-2.0.5-15.el9.x86_64.rpm	1.8 MB/s 337 kB	00:00
(2/6): kea-libs-2.6.4-1.el9.x86_64.rpm	4.9 MB/s 3.0 MB	00:00
(3/6): mariadb-connector-c-3.2.6-1.el9_0.x86_64.rpm	443 kB/s 195 kB	00:00
(4/6): kea-2.6.4-1.el9.x86_64.rpm	1.9 MB/s 1.3 MB	00:00
(5/6): mariadb-connector-c-config-3.2.6-1.el9_0.noarch.rpm	59 kB/s 9.8 kB	00:00
(6/6): postgresql-private-libs-13.23-1.el9_7.x86_64.rpm	868 kB/s 140 kB	00:00

```
=====
```

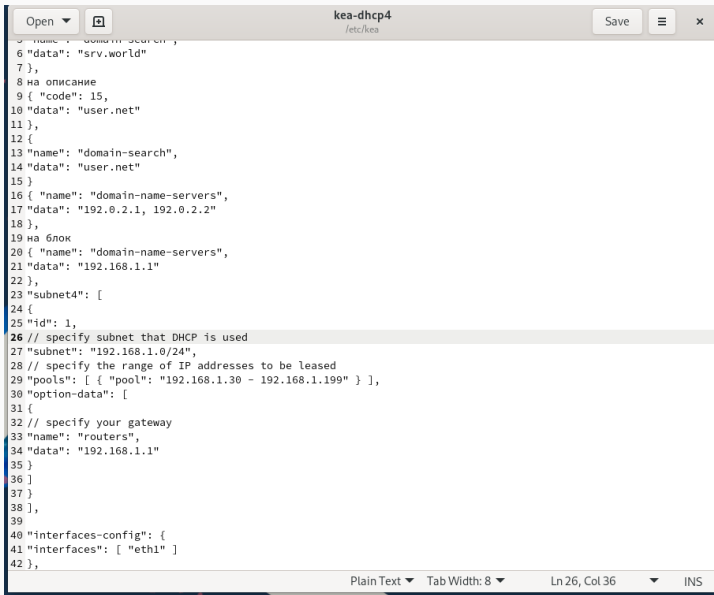
Total	42 kB/s 5.0 MB	02:01
-------	------------------	-------

```
=====
```

Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction

Preparing	:	1/1
Installing	: postgresql-private-libs-13.23-1.el9_7.x86_64	1/6
Installing	: mariadb-connector-c-config-3.2.6-1.el9_0.noarch	2/6
Installing	: mariadb-connector-c-3.2.6-1.el9_0.x86_64	3/6

Настройка сетевых параметров и DNS-опций



The screenshot shows a text editor window titled 'kea-dhcp4' with a file icon and 'etc/kea' path. The editor contains a JSON configuration for a DHCP server. The configuration includes domain search settings, domain name servers, and a subnet configuration with a pool of IP addresses and a gateway. The status bar at the bottom indicates 'Plain Text', 'Tab Width: 8', 'Ln 26, Col 36', and 'INS'.

```
1 { "name": "domain-search",
2   "data": "srv.world"
3 },
4 на описание
5 { "code": 15,
6   "data": "user.net"
7 },
8 {
9   "name": "domain-search",
10  "data": "user.net"
11 },
12 {
13   "name": "domain-name-servers",
14   "data": "192.0.2.1, 192.0.2.2"
15 },
16 на блок
17 { "name": "domain-name-servers",
18   "data": "192.168.1.1"
19 },
20 "subnet4": [
21 {
22   "id": 1,
23   // specify subnet that DHCP is used
24   "subnet": "192.168.1.0/24",
25   // specify the range of IP addresses to be leased
26   "pools": [ { "pool": "192.168.1.30 - 192.168.1.199" } ],
27   "option-data": [
28     {
29       // specify your gateway
30       "name": "routers",
31       "data": "192.168.1.1"
32     }
33 ]
34 },
35 ],
36 "interfaces-config": {
37   "interfaces": [ "eth1" ]
38 },
39 }
```

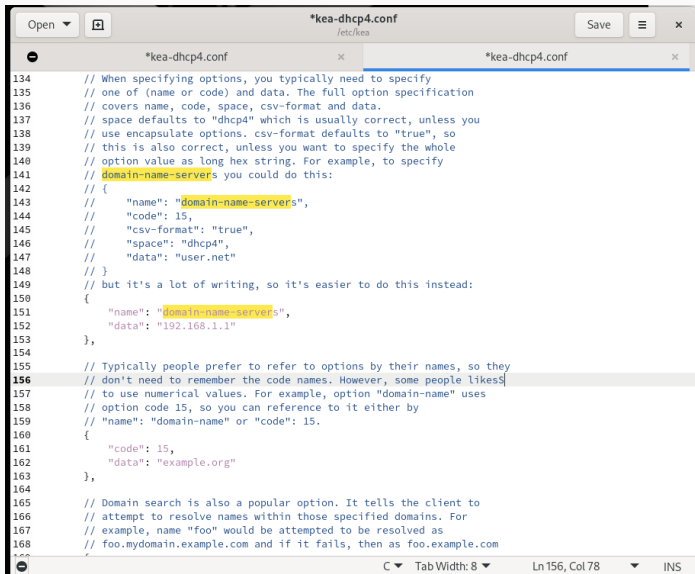
Plain Text ▾ Tab Width: 8 ▾ Ln 26, Col 36 ▾ INS

Привязка к интерфейсу и проверка конфигурации

```
[ 44.621644] vmagfx 0000:00:02.0: [drm] DMA map mode: Caching DMA mappings.
[ 44.624452] vmagfx 0000:00:02.0: [drm] Legacy memory limits: VRAM = 262144 kB
, FIFO = 2048 kB, surface = 262144 kB
[ 44.624462] vmagfx 0000:00:02.0: [drm] MOB limits: max mob size = 131072 kB,
max mob pages = 262144
[ 44.624467] vmagfx 0000:00:02.0: [drm] Max GMR ids is 8192
[ 44.624470] vmagfx 0000:00:02.0: [drm] Max number of GMR pages is 1048576
[ 44.624472] vmagfx 0000:00:02.0: [drm] Maximum display memory size is 262144
kB
[ 44.743257] vmagfx 0000:00:02.0: [drm] Screen Target display unit initialized
[ 44.749736] vmagfx 0000:00:02.0: [drm] Fifo max 0x00200000 min 0x00001000 cap
0x00000355
[ 44.750680] vmagfx 0000:00:02.0: [drm] Using command buffers with DMA pool.
[ 44.750704] vmagfx 0000:00:02.0: [drm] Available shader model: SM_5.
[ 44.754230] [drm] Initialized vmagfx 2.20.0 20211206 for 0000:00:02.0 on mino
r 0
[ 44.770198] fbcon: vmagfxdrmfb (fb0) is primary device
[ 44.786079] Console: switching to colour frame buffer device 160x50
[ 44.800055] vmagfx 0000:00:02.0: [drm] fb0: vmagfxdrmfb frame buffer device
[ 44.813079] snd_intel8x0 0000:00:05.0: intel8x0_measure_ac97_clock: measured 56364 usecs (2640 samples)
[ 44.813091] snd_intel8x0 0000:00:05.0: clocking to 49190
[ 50.307040] systemd-rc-local-generator[7391]: /etc/rc.d/rc.local is not marked executable, skipping.
[ 53.105405] NET: Registered PF_QIPCRTR protocol family
[ 58.765459] e1000: eth0 NIC Link is Up 1000 Mbps Full Duplex, Flow Control: RX
[ 58.800401] IPv6: ADDRCONF(NETDEV_CHANGE): eth0: link becomes ready
[ 58.900760] e1000: eth1 NIC Link is Up 1000 Mbps Full Duplex, Flow Control: RX
[ 58.965799] Warning: Unmaintained driver is detected: ip_set
[ 59.814644] IPv6: ADDRCONF(NETDEV_CHANGE): eth1: link becomes ready
[ 91.092689] block sda: the capability attribute has been deprecated.
```

Рис. 3: Проверка конфигурационного файла

Интеграция с DNS: прямые и обратные зоны



```
134 // When specifying options, you typically need to specify
135 // one of (name or code) and data. The full option specification
136 // covers name, code, space, csv-format and data.
137 // space defaults to "dhcp4" which is usually correct, unless you
138 // use encapsulate options. csv-format defaults to "true", so
139 // this is also correct, unless you want to specify the whole
140 // option value as long hex string. For example, to specify
141 // domain-name-servers you could do this:
142 // {
143 //     "name": "domain-name-servers",
144 //     "code": 15,
145 //     "csv-format": "true",
146 //     "space": "dhcp4",
147 //     "data": "user.net"
148 // }
149 // but it's a lot of writing, so it's easier to do this instead:
150 {
151     "name": "domain-name-servers",
152     "data": "192.168.1.1"
153 },
154
155 // Typically people prefer to refer to options by their names, so they
156 // don't need to remember the code names. However, some people likes
157 // to use numerical values. For example, option "domain-name" uses
158 // option code 15, so you can reference to it either by
159 // "name": "domain-name" or "code": 15.
160 {
161     "code": 15,
162     "data": "example.org"
163 },
164
165 // Domain search is also a popular option. It tells the client to
166 // attempt to resolve names within those specified domains. For
167 // example, name "foo" would be attempted to be resolved as
168 // foo.mydomain.example.com and if it fails, then as foo.example.com
```


Обратная DNS-зона

```
"subnet4": [
{
    // This defines the whole subnet. Kea will use this information to
    // determine where the clients are connected. This is the whole
    // subnet in your network.

    // Subnet identifier should be unique for each subnet.
    "id": 1,

    // This is mandatory parameter for each subnet.
    "subnet": "192.168.1.0/24",

    // Pools define the actual part of your subnet that is governed
    // by Kea. Technically this is optional parameter, but it's
    // almost always needed for DHCP to do its job. If you omit it,
    // clients won't be able to get addresses, unless there are
    // host reservations defined for them.
    "pools": [ { "pool": "192.168.1.30 - 192.168.1.199" } ],

    // These are options that are subnet specific. In most cases,
    // you need to define at least routers option, as without this
    // option your clients will not be able to reach their default
    // gateway and will not have Internet connectivity.
    "option-data": [
        {
            // For each IPv4 subnet you most likely need to specify at
            // least one router.
            "name": "routers",
            "data": "192.168.1.1"
        }
    ],

    // Kea offers host reservations mechanism. Kea supports reservations
    // by several different types of identifiers: hw-address
    // (hardware/MAC address of the client), duid (DUID inserted by the
    // client), client-id (client identifier inserted by the client) and
    // circuit-id (circuit identifier inserted by the relay agent).
    //
```

Проверка доступности DHCP-сервера по имени

Остальные примеры задания конфигураций подсетей удалите.

3. Настройте привязку dhcpd к интерфейсу eth1 виртуальной машины server:

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```
"interfaces-config": {  
  "interfaces": [ "eth1" ]  
},
```

4. Проверьте правильность конфигурационного файла:

```
kea-dhcp4 -t /etc/kea/kea-dhcp4.conf
```

Рис. 6: Проверка доступности имени dhcp.akabrisov.net

SELinux и межсетевой экран

```
23 ## Server configuration
24 config.vm.define "server", autostart: false do |server|
25   server.vm.box = "rocky9"
26   server.vm.hostname = 'server'
27   server.vm.synced_folder ".", "/vagrant", disabled: true
28
29   server.vm.boot_timeout = 1440
30
31   server.ssh.insert_key = false
32   server.ssh.username = 'vagrant'
33   server.ssh.password = 'vagrant'
34
35   server.vm.network :private_network,
36     ip: "192.168.1.1",
37     virtualbox____intnet: true
38
39   server.vm.provision "server dummy",
40     type: "shell",
41     preserve_order: true,
42     path: "provision/server/01-dummy.sh"
43   server.vm.provision "server dns"
```

Рис. 7: Настройка SELinux и firewall

Получение адреса клиентом по DHCP

```
26 // DHCPv4 configuration starts here. This section will be read by DHCPv4 server
27 // and will be ignored by other components.
28 "Dhcp4": {
29     // Add names of your network interfaces to listen on.
30     "interfaces-config": {
31         // See section 8.2.4 for more details. You probably want to add just
32         // interface name (e.g. "eth0" or specific IPv4 address on that
33         // interface name (e.g. "eth0/192.0.2.1").
34         "interfaces": ["eth1" ]
35     }
36     // Kea DHCPv4 server by default listens using raw sockets. This ensures
37     // all packets, including those sent by directly connected clients
38     // that don't have IPv4 address yet, are received. However, if your
39     // traffic is always relayed, it is often better to use regular
40     // UDP sockets. If you want to do that, uncomment this line:
41     // "dhcp-socket-type": "udp"
42 },
43
44 // Kea supports control channel, which is a way to receive management
45 // commands while the server is running. This is a Unix domain socket that
46 // receives commands formatted in JSON, e.g. config-set (which sets new
47 // configuration), config-reload (which tells Kea to reload its
48 // configuration from file), statistic-get (to retrieve statistics) and many
49 // more. For detailed description, see Sections 8.8, 16 and 15.
50 "control-socket": {
51     "socket-type": "unix",
52     "socket-name": "kea4-ctrl-socket"
53 },
54
55 // Use Memfile lease database backend to store leases in a CSV file.
56 // Depending on how Kea was compiled, it may also support SQL databases
```

Параметры сетевых интерфейсов клиента

```
[root@server ~]# kea-dhcp4 -t /etc/kea/kea-dhcp4.conf
2026-03-04 19:52:51.138 INFO [kea-dhcp4.hosts/27065.139698838464640] HOSTS_BACKENDS_REGISTER
ED the following host backend types are available: mysql postgresql
2026-03-04 19:52:51.144 WARN [kea-dhcp4.dhcp4/27065.139698838464640] DHCP4_RESERVED_Q
UEUE_CONTROL disabling dhcp queue control when multi-threading is enabled.
2026-03-04 19:52:51.145 WARN [kea-dhcp4.dhcp4/27065.139698838464640] DHCP4_RESERVED_LOOK
UP_FIRST_ENABLED Multi-threading is enabled and host reservations lookup is always performed
first.
2026-03-04 19:52:51.146 INFO [kea-dhcp4.dhcp4/27065.139698838464640] DHCP4_RESERVED_NEW_SU
BNET4 a new subnet has been added to configuration: 192.168.1.0/24 with params: t1=900, t2=18
00, valid-lifetime=3600
2026-03-04 19:52:51.146 ERROR [kea-dhcp4.dhcp4/27065.139698838464640] DHCP4_RESERVED_FAIL faile
d to create or run parser for configuration element subnet4: specified reservation '192.0.2.2
01' is not within the IPv4 subnet '192.168.1.0/24'
Error encountered: specified reservation '192.0.2.201' is not within the IPv4 subnet '192.168
.1.0/24'
[root@server ~]# systemctl --system daemon-reload
[root@server ~]# systemctl enable kea-dhcp4.service
Created symlink /etc/systemd/system/multi-user.target.wants/kea-dhcp4.service → /usr/lib/syst
emd/system/kea-dhcp4.service.
[root@server ~]#
```

Рис. 9: Вывод ifconfig на клиенте

Создание и подключение TSIG-ключа

```
(gedit:27131): dconf-WARNING **: 19:54:00.846: failed to commit changes to dconf: Failed to execute child process "dbus-launch" (No such file or directory)
[root@server ~]# /var/named/master/fz
-bash: /var/named/master/fz: No such file or directory
[root@server ~]# cd ..
[root@server /]# cd /var/named/master/fz
-bash: cd: /var/named/master/fz: No such file or directory
[root@server /]# cd /var
[root@server var]# cd master
-bash: cd: master: No such file or directory
[root@server var]# systemctl restart named
Failed to restart named.service: Unit named.service not found.
[root@server var]# ping dhcp.user.net
ping: dhcp.user.net: Name or service not known
[root@server var]# chown -R named:named /var/named
chown: invalid user: 'named:named'
[root@server var]#
```

Рис. 10: Создание ключа TSIG

Разрешение обновлений DNS-зон (Bind)

```
root@server:/var
cockpit dhcpv6-client imap imaps pop3 pop3s smtp ssh
[root@server var]# firewall-cmd --get-services
firewall-cmd --add-service=dhcp
firewall-cmd --add-service=dhcp --permanent
RH-Satellite-6 RH-Satellite-6-capsule afp amanda-client amanda-k5-client amqp amqps apcupsd a
udit ausweisapp2 bacula bacula-client bareos-director bareos-filedaemon bareos-storage bb bgp
bitcoin bitcoin-rpc bitcoin-testnet bitcoin-testnet-rpc bittorrent-lsd ceph ceph-exporter ce
ph-mon cfengine checkmk-agent cockpit collectd condor-collector cratedb ctdb dds dds-multicas
t dds-unicast dhcp dhcpv6 dhcpv6-client distcc dns dns-over-tls docker-registry docker-swarm
dropbox-lansync elasticsearch etcd-client etcd-server finger foreman foreman-proxy freeipa-4
freeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp galera ganglia-client gangli
a-master git gpsd grafana gre high-availability http http3 https ident imap imaps ipfs ipp ip
p-client ipsec irc ircs iscsi-target isns jenkins kadmin kdeconnect kerberos kibana klogind kp
asswd kprop kshell kube-api kube-apiserver kube-control-plane kube-control-plane-secure kube-
controller-manager kube-controller-manager-secure kube-nodeport-services kube-scheduler kube-
scheduler-secure kube-worker kubelet kubelet-readonly kubelet-worker ldap ldaps libvirt libvi
rt-tls lightning-network llmnr llmnr-client llmnr-tcp llmnr-udp managesieve matrix mdns memca
che minidlna mongodb mosh mountd mqtt mqtt-tls ms-wbt mssql murmur mysql nbd nebula netbios-n
s netdata-dashboard nfs nfs3 nmea-0183 nrpe ntp nut openvpn ovirt-imageio ovirt-storageconsol
e ovirt-vmconsole plex pmcd pmproxy pmwebapi pmwebapis pop3 pop3s postgresql privoxy promethe
us prometheus-node-exporter proxy-dhcp ps2link ps3netshr ptp pulseaudio puppetmaster quassel
radius rdp redis redis-sentinel rpc-bind rquotad rsh rsyncd rtsp salt-master samba samba-clie
nt samba-dc sane sip sips slp smtp smtp-submission smtps snmp snmptls snmptls-trap snmptrap s
piderOak-lansync spotify-sync squid ssdp ssh steam-streaming svdrp svn syncthing syncthing-gu
i syncthing-relay synergy syslog syslog-tls telnet tentacle tftp tile38 tinc tor-socks transm
ission-client upnp-client vdsm vnc-server warpinator wbem-http wbem-https wireguard ws-discov
ery ws-discovery-client ws-discovery-tcp ws-discovery-udp wsman wsmans xdmcp xmpp-bosh xmpp-c
lient xmpp-local xmpp-server zabbix-agent zabbix-server zerotier
success
success
[root@server var]#
```

Описание TSIG-ключа для Kea

```
us prometheus-node-exporter proxy-dhcp ps2link ps3netsrv ptp pulseaudio puppetmaster quassel
radius rdp redis redis-sentinel rpc-bind rquotad rsh rsyncd rtsp salt-master samba samba-clie
nt samba-dc sane sip sips slp smtp smtp-submission smtps snmp snmptls snmptls-trap snmptrap s
pideroak-lansync spotify-sync squid ssdp ssh steam-streaming svdrp svn syncthing syncthing-gu
i syncthing-relay synergy syslog syslog-tls telnet tentacle tftp tile38 tinc tor-socks transm
ission-client upnp-client vdsm vnc-server warpinator wbem-http wbem-https wireguard ws-discov
ery ws-discovery-client ws-discovery-tcp ws-discovery-udp wsman wsmans xdmcp xmpp-bosh xmpp-c
lient xmpp-local xmpp-server zabbix-agent zabbix-server zerotier
success
success
[root@server var]# estorecon -vR /etc
bash: estorecon: command not found...
[root@server var]# restorecon -vR /etc
[root@server var]# restorecon -vR /var/named
restorecon: lstat(/var/named) failed: No such file or directory
[root@server var]# restorecon -vR /var/lib/kea/
[root@server var]# tail -f /var/log/messages
Mar  4 19:53:10 localhost systemd-rc-local-generator: /etc/rc.d/rc.local is not marked execut
able, skipping.
Mar  4 19:53:17 localhost systemd[1]: Reloading.
Mar  4 19:53:17 localhost systemd-rc-local-generator: /etc/rc.d/rc.local is not marked execut
able, skipping.
Mar  4 19:53:57 localhost gnome-shell[24477]: Window manager warning: Buggy client sent a _NE
T_ACTIVE_WINDOW message with a timestamp of 0 for 0x30000f8
Mar  4 19:53:57 localhost systemd[1]: Starting dnf makecache...
Mar  4 19:53:57 localhost dnf[27149]: Metadata cache refreshed recently.
Mar  4 19:53:58 localhost systemd[1]: dnf-makecache.service: Deactivated successfully.
Mar  4 19:53:58 localhost systemd[1]: Finished dnf makecache.
Mar  4 19:58:34 localhost systemd[1]: Starting PackageKit Daemon...
Mar  4 19:58:34 localhost systemd[1]: Started PackageKit Daemon.
```


Настройка kea-dhcp-ddns



```
*01-routing.sh
1 echo "Provisioning script $0"
2 nmcli connection modify "eth1" ipv4.gateway "192.168.1.1"
3 nmcli connection up "eth1"
4 nmcli connection modify eth0 ipv4.never-default true
5 nmcli connection modify eth0 ipv6.never-default true
6 nmcli connection down eth0
7 nmcli connection up eth0
```

Рис. 13: Файл kea-dhcp-ddns.conf

Запуск службы DHCP-DDNS

```
xecute child process "dbus-launch" (No such file or directory)
[root@server /]# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fe80::a00:27ff:fe69:a8d prefixlen 64 scopeid 0x20<link>
    inet6 fd17:625c:f037:2:a00:27ff:fe69:a8d prefixlen 64 scopeid 0x0<global>
    ether 08:00:27:69:0a:8d txqueuelen 1000 (Ethernet)
    RX packets 1479236 bytes 2019630895 (1.8 GiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 178954 bytes 12543603 (11.9 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.1.1 netmask 255.255.255.0 broadcast 192.168.1.255
    inet6 fe80::a00:27ff:fee3:341f prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:e3:34:1f txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 44 bytes 5322 (5.1 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 17 bytes 2039 (1.9 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 17 bytes 2039 (1.9 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

[root@server /]#
```

Включение DDNS в kea-dhcp4.conf

```
[root@server /]# cat /var/lib/kea/kea-leases4.csv  
[root@server /]# mkdir -p /etc/named/keys  
[root@server /]# mkdir -p /etc/named/keys  
tsig-keygen -a HMAC-SHA512 DHCP_UPDATER > /etc/named/keys/dhcp_updater.key  
[root@server /]# gedit /etc/named/keys/dhcp_updater.key  
  
(gedit:27344): dconf-WARNING **: 20:12:32.361: failed to commit changes to dconf: Fa
```

Рис. 15: Фрагмент обновлённого kea-dhcp4.conf

Запуск и проверка службы Kea DHCP

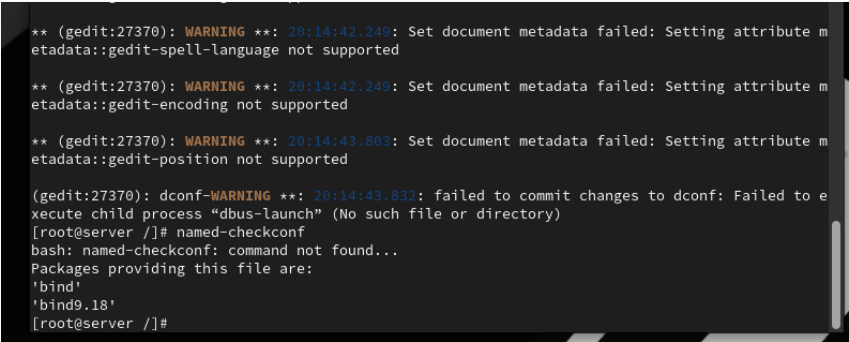


The image shows a text editor window titled "dhcp_updater.key" with the path "/etc/kea/keys". The editor contains the following JSON configuration:

```
1 key "DHCP_UPDATER" {  
2     algorithm hmac-sha512;  
3     secret "gy3Cy8Pe3eEsw7TmjEjJf1/K7RzFMSn5bw2HZ0XMhu7NUQNd1017KDNI2ye5ZKJ1yytiWY15jKv/hwvi7ig0rg==";  
4 };
```

Рис. 16: Запуск kea-dhcp

Проверка создания DNS-записей для клиента

A terminal window with a dark background and light-colored text. It shows several warning messages from gedit and an error from the dig command. The warnings are about unsupported metadata attributes: gedit-spell-language, gedit-encoding, and gedit-position. The error from dig is about a failed commit to dconf and a missing dbus-launch process. The terminal prompt is [root@server /]#.

```
** (gedit:27370): WARNING **: 20:14:42.249: Set document metadata failed: Setting attribute metadata::gedit-spell-language not supported

** (gedit:27370): WARNING **: 20:14:42.249: Set document metadata failed: Setting attribute metadata::gedit-encoding not supported

** (gedit:27370): WARNING **: 20:14:43.803: Set document metadata failed: Setting attribute metadata::gedit-position not supported

(gedit:27370): dconf-WARNING **: 20:14:43.832: failed to commit changes to dconf: Failed to execute child process "dbus-launch" (No such file or directory)
[root@server /]# named-checkconf
bash: named-checkconf: command not found...
Packages providing this file are:
'bind'
'bind9.18'
[root@server /]#
```

Рис. 17: Результат выполнения dig

Интерпретация результата dig

- Заголовок: успешный ответ (NOERROR)
- Сервер ответа: 192.168.1.1 (server)
- QUESTION: client.akabrisov.net
- ANSWER: IP-адрес клиента из DHCP
- Вывод: DHCP → Kea DDNS → Bind9 работают согласованно

Скрипт автоматической настройки dhcpr.sh

A screenshot of a text editor window titled 'tsig-keys.json' with the path '/etc/kea' below it. The window has a toolbar with 'Open', 'Save', and a menu icon. The code is as follows:

```
1 "tsig-keys" : [  
2 {  
3   "name": "DHCP_UPDATER",  
4   "algorithm": "hmac-sha512",  
5   "secret":  
6     "psFFSGdUIwK36l1G82L0w15PuIH4W5uB8h/cw7F0XDsjniBbwm/59tM+V3ydcCs15VLpe2pxlUlg"  
7 }  
8 ]
```

Рис. 18: Скрипт dhcpr.sh для автоматической настройки

Выводы по проделанной работе

- Настроены сервисы:
 - Kea DHCP (выдача IP-адресов)
 - Bind9 DNS (прямые и обратные зоны)
- Реализовано динамическое обновление DNS-зон (DDNS) с использованием TSIG-ключей и Kea DHCP-DDNS.